

WEEK 4 – POWER BI EDA

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INTRODUCTION:

Exploratory Data Analysis (EDA) is the process of studying and understanding data using calculations and visualizations. In this project, Power BI Desktop is used to analyze stock data.

The dashboard helps analyze stock prices, trading volume, moving averages, and daily price changes. Interactive slicers and a custom tooltip are also used to make the dashboard easier to explore.

OBJECTIVES:

The main objectives of this project are:

- To analyze stock price data.
- To calculate 20-day and 50-day Simple Moving Averages.
- To compare moving averages using a crossover signal.
- To analyze trading volume.
- To create an interactive stock price dashboard.
- To provide different price metric selections.
- To use date slicers for interactive analysis.
- To display detailed stock information using a custom tooltip.

DATA ANALYSIS:

The project uses DAX measures to calculate moving averages based on the stock trading volume.

4.1 Volume 20-Day Simple Moving Average

The 20-day SMA calculates the average trading volume over the previous 20 days.

Volume 20 SMA = VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 20

VAR DateInPeriod = DATESINPERIOD('Date Table'[Date], CurrentDate, -
PeriodDays, DAY)

RETURN CALCULATE(AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]),
DateInPeriod)

4.2 Volume 50-Day Simple Moving Average

The 50-day SMA calculates the average trading volume over the previous 50 days.

Volume 50 SMA =

VAR CurrentDate = MAX('Date Table'[Date])

VAR PeriodDays = 50

VAR DateInPeriod = DATESINPERIOD('Date Table'[Date], CurrentDate, -
PeriodDays, DAY)

RETURN

CALCULATE(

AVERAGE('SHOP_2015-05-21_2025-03-16'[Volume]), DateInPeriod)

These moving averages help compare recent trading volume with longer-term volume trends.

Moving Average Crossover:

A moving average crossover signal is created to compare the 20-day and 50-day moving averages.

MA Crossover Signal =

IF(

ISBLANK([Volume 20 SMA]) ||

ISBLANK([Volume 50 SMA]),

BLANK(),

```

IF(
    [Volume 20 SMA] > [Volume 50 SMA],
    "Bullish (20 > 50)",
    "Bearish (20 < 50)"
)
)

```

The measure displays a Bullish signal when the 20-day SMA is greater than the 50-day SMA and a Bearish signal otherwise.

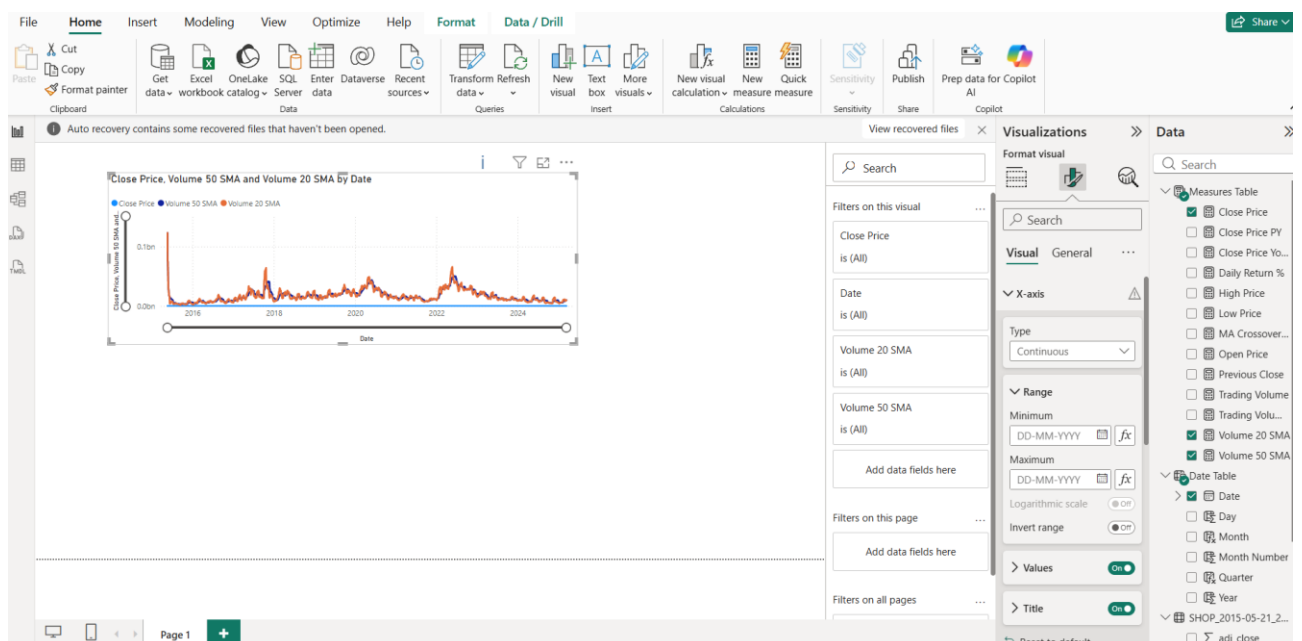
PRICE TREND VISUALIZATIONS:

An interactive line and stacked column chart is used to analyze the stock price trend.

The chart contains:

- Date on the X-axis
- Close Price
- Volume 20 SMA
- Volume 50 SMA

The date axis is used as a continuous timeline. A zoom slider is also enabled so that users can examine different periods of the stock data more easily.



TRADING VOLUME ANALYSIS:

A separate chart is created to analyze trading volume.

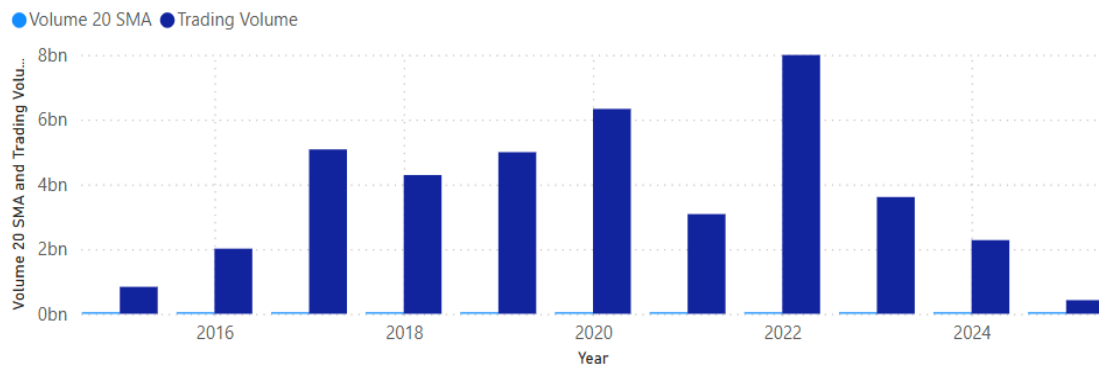
The chart contains:

- Date/Year on the X-axis
- Trading Volume
- Volume 20 SMA

This visualization helps compare actual trading volume with the 20-day average volume.

Large differences between trading volume and the moving average can help identify periods of increased trading activity.

Volume 20 SMA and Trading Volume by Year



VOLUME COLOR MEASURE:

A DAX measure was created to identify volume spikes.

Volume Color =

```
IF(  
    [Trading Volume] > [Volume 20 SMA] * 1.5,  
    "#8E44AD",  
    "#2E86C1"  
)
```

The measure identifies volume values that are more than 1.5 times the 20-day moving average. The report specifies red for such volume spikes and green for other values.

DATE RANGE SLICER:

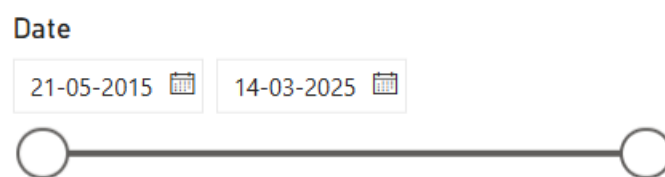
A date range slicer is added to the dashboard.

The slicer allows the user to select a specific start and end date. This makes it possible to analyze stock information for a selected time period.

The dashboard contains a date range from:

21-05-2015 to 14-03-2025

as shown in the completed dashboard.



PRICE METRIC SELECTOR:

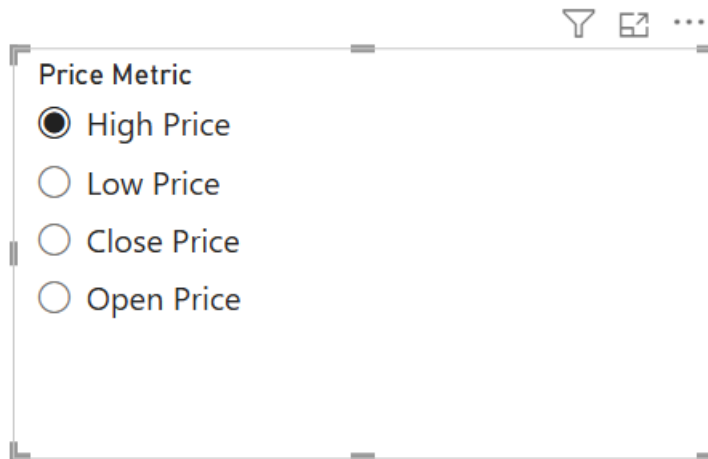
A dynamic Price Metric selector is created using a field parameter.

The available options are:

- High Price
- Low Price
- Close Price
- Open Price

The user can select any one of these metrics and analyze it dynamically in the dashboard.

In the completed dashboard, Close Price is selected.



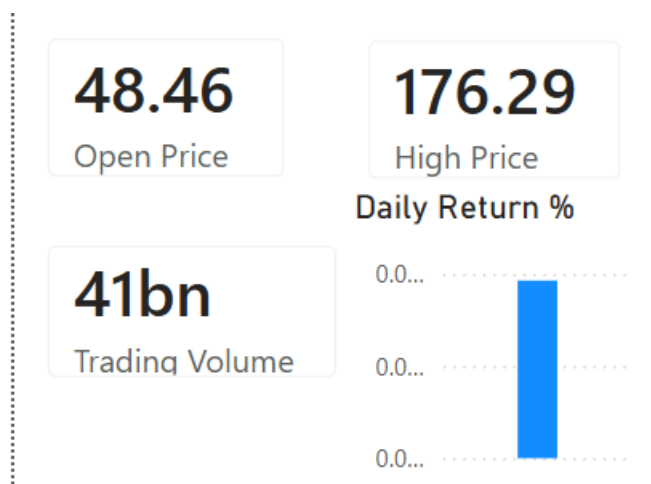
CUSTOM TOOLTIP:

A separate report page named Stock_Tooltip is created for displaying detailed information when the user moves the mouse over the main chart.

The tooltip contains:

- Open Price
- High Price
- Low Price
- Close Price
- Trading Volume
- Daily Change %

The page is configured as a tooltip page and connected to the main chart.



DASHBOARD:

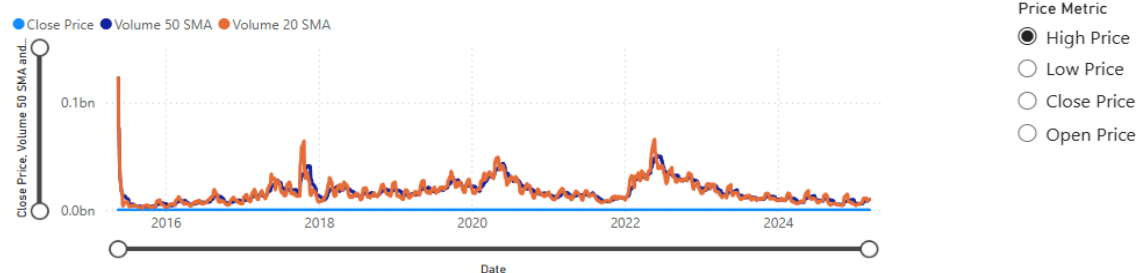
The dashboard provides several interactive features:

- Date range selection
- Relative date selection
- Price metric selection
- Zoom slider
- Interactive price chart
- Trading volume analysis
- Custom tooltip

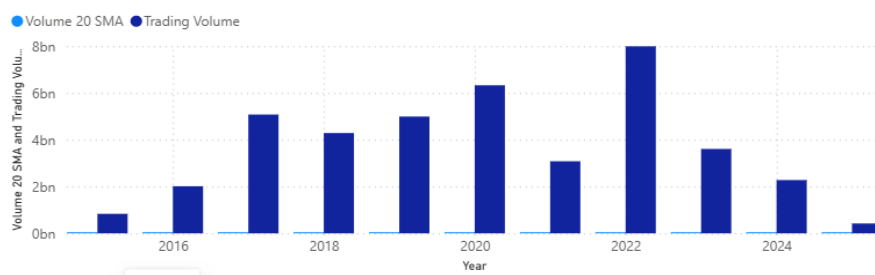
These features allow users to explore the stock data according to their selected time period and price metric.

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Close Price, Volume 50 SMA and Volume 20 SMA by Date



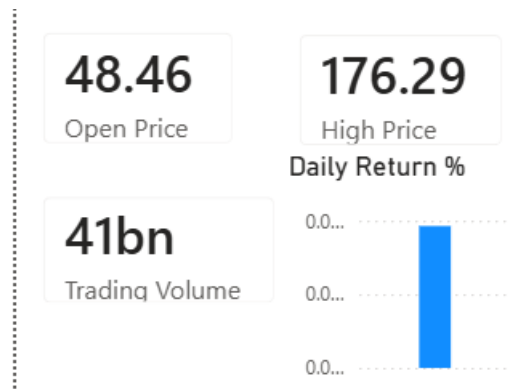
Volume 20 SMA and Trading Volume by Year



PAGE 2:



PAGE 3:



CONCLUSION:

The Week 4 Power BI EDA project provides an interactive dashboard for analyzing Shopify stock data. DAX was used to calculate 20-day and 50-day moving averages and create a moving average crossover signal.

The dashboard combines stock price trends, trading volume, slicers, a dynamic price selector, KPI cards, and a custom tooltip. These features make the stock dataset easier to explore and understand using Power BI Desktop.